KILNDECK · TRACEABILITY MATRIX

KilnDeck traceability matrix

Design Molecule Lab · designmolecule.com

DOCUMENT CONTROL

Document ID	KND-RTM-001	Version	0.4
Status	Draft	Date	2026-10-30
Author	Amish Chadha	License	CERN-OHL-S-2.0
Project TRL	TRL 3 Proof of concept		
Repository	github.com/BoujeeEnjinia1701/kilndeck		
Source	docs/03b-traceability.md @ 9f92bd8		

REVISION HISTORY

Version	Date	Author	Change
0.4	2026-10-30	Amish Chadha	Generated by .kit/trace.py
0.3	2026-10-30	Amish Chadha	Generated by .kit/trace.py
0.2	2026-10-29	Amish Chadha	Generated by .kit/trace.py
0.1	2026-10-29	Amish Chadha	Generated by .kit/trace.py

KilnDeck traceability matrix

Generated by `python .kit/trace.py` from the user requirements, product requirements, calculations, build plan and BOM. Do not edit by hand. 14 user requirements, 17 product requirements, 0 gaps.

Table 1. User requirement to product requirement to evidence to verification.

User requirement	Product requirement	Title	Evidence	Verification	Status
UR-001, UR-003	PR-001	No steps on the crust	KND-CAL-001, KND-BLD-001, KND-OPT-001	Observed trial over 20 rounds, TRL 4	Met on the straight sides; not met on the curved ends, which this prototype does not serve
UR-002	PR-002	Guardrail	KND-CAL-001, KND-BLD-001, KND-STM-001	Analysis (KND-CAL-001 s7), TRL 3; proof load (CalRig), TRL 4	Met on paper: 150 MPa at the proof load (yield 275 MPa)
UR-002	PR-003	Rated load	KND-CAL-001, KND-BLD-001, KND-STM-001	Analysis (KND-CAL-001 s6), TRL 3; proof load, TRL 4	Met on paper: 21 MPa in the chords at the proof load, with the carriage at mid-span
UR-004	PR-004	Hand wheel force	KND-CAL-001, KND-BLD-001, KND-STM-001	Analysis (KND-CAL-001 s10), TRL 3; force gauge, TRL 4	Met on paper: 68 N level, 135 N on a 2 % grade
UR-006	PR-005	Fuel dose	KND-CAL-001, KND-BLD-001, KND-STM-001	Weigh 20 doses, TRL 4	To verify at TRL 4; nominal 0.49 kg of coal per pocket
UR-003	PR-006	Hopper capacity	KND-CAL-001, KND-STM-001	Analysis (KND-CAL-001 s3), TRL 3	Met on paper: 52 L, about 42 kg of coal against about 30 kg a round (1.4 rounds)
UR-005	PR-007	Floor temperature	KND-CAL-001, KND-BLD-001, KND-STM-001, KND-OPT-001	Analysis (KND-CAL-001 s11), TRL 3; contact thermometer log, TRL 4	Not met in full sun: about 74 °C at 45 °C air; met at night (34 °C) and in shade (48 °C); the bare crust is about 150 °C
UR-007	PR-008	Load on the kiln	KND-CAL-001, KND-BLD-001, KND-STM-001	Analysis (KND-CAL-001 s9), TRL 3; survey and load cell, TRL 4	Met against the interim 0.20 MPa (0.154 MPa; 0.209 MPa at the 250 kg rated load); cannot be confirmed until the survey
UR-003	PR-009	Spout alignment	KND-CAL-001, KND-BLD-001	Analysis (KND-CAL-001 s12), TRL 3; measure at 10 holes, TRL 4	Met on paper: about 17 mm along the kiln, 10 mm across
UR-013	PR-010	Parking brake	KND-CAL-001, KND-BLD-001, KND-STM-001	Analysis (KND-CAL-001 s10), TRL 3; inclined track test, TRL 4	Met on paper: holds 767 N against 512 N
UR-010	PR-011	Cost	KND-CAL-001	Costed BOM, TRL 3	Estimated cost of the constructable design: USD 2,905 (USD 595 under the target)
UR-008	PR-012	Piece mass		Analysis from the model solids (KND-	Not met**: the end truss modules weigh 34.4 and 34.8 kg (1.4 and 1.8

User requirement	Product requirement	Title	Evidence	Verification	Status
			KND-CAL-001, KND-BLD-001, KND-STM-001	CAL-001 s5), TRL 3; weigh, TRL 4	kg over); every other piece is inside 33 kg (middle truss modules 32.9 kg, outer end truck 29.8 kg, track modules 26.7 kg, floor panels 24.8 kg); ways to close it in KND-DEC-001 D16
UR-013	PR-013	Track retention	KND-CAL-001, KND-BLD-001	Analysis (KND-CAL-001 s9, s17), TRL 3; push and lift tests, TRL 4	Met by design: end stops, buffers and derailment guards; the stops take 4.2 kN at a running speed of 0.28 m/s
UR-012, UR-013	PR-014	Storm hold	KND-CAL-001	Analysis (KND-CAL-001 s15), TRL 3; pull test on the clamps and anchors, TRL 4	Met on paper with the storm clamps and tie-downs (KND-DEC-001 D12); anchor pull-out to assess at the partner kiln
UR-011	PR-015	Local build	KND-CAL-001	Inspection of the model and build plan (KND-BLD-001), TRL 3	Met by design
UR-009	PR-016	Track relaying	KND-CAL-001	Inspection (KND-BLD-001), TRL 3; relay trial, TRL 4	Met by design
UR-014	PR-017	Upkeep	KND-CAL-001	Inspection of the BOM and model, TRL 3	Met by design

Gaps

None.